Practical Deep Reinforcement Learning Approach for Stock Trading Implementation

Project Report
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1 Introduction

In this project the we study and attempt to replicate a stock trading strategy with Deep Reinforcement Learning approach proposed by Liu et al [3]. The author mentioned two approaches for seeking profitable trading strategies, one through the traditional method of optimizing portfolio weights and the other through modeling the stock trading problem as a Markov Decision Process (MDP) and apply Dynamic Programming techniques. The traditional techniques such as value iteration are often computationally inapplicable when state and action spaces are large and deep reinforcement learning techniques are used for approximating the solutions. Liu et al explored the Deep Deterministic Policy Gradient algorithm (DDPG), which uses the actor-critic framework to model large state and action spaces, target network to stabilize the training process, and experience replay for correlation reduction between samples, and claim that the DDPG algorithm over performs the mean-variance portfolio and DJIA. In this report, we will give a general overview of the algorithm, then we will explain our attempt in replicating the strategy, finally we present the obtained numerical results.

2 An Overview of the Deep Deterministic Policy Gradient (DDPG) Algorithm

2.1 Trading problem as a Markov decision process

Based on [3], the stock trading process can be formulated as a Markov Decision Process (MDP). Let $\mathbb{R}$ denote the set of real numbers, let $\mathbb{R}_+ = \{x \in \mathbb{R} : x \geq 0\}$ denote the set of nonnegative real numbers. Let n be the number of stocks in the market. The agent observes the state

$$s = [p, h, b],$$

where $p \in \mathbb{R}_+^n$ denotes the vector of stock prices, $h \in \mathbb{R}_+^n$ denotes the vector of stock holdings, and $b \in \mathbb{R}_+$ denotes the remaining cash balance.

The action vector

$$a = (a_1, \ldots, a_n)$$

specifies the trading decision for each of the n stocks.

For $i = 1, \ldots$,n, each component a_i indicates whether to buy, sell, or hold stock i, thereby increasing, decreasing, or leaving unchanged the corresponding holding h_i, respectively.

The reward function $r(s, a, s')$ is defined as the change in portfolio value when action a is taken at state s and the environment transitions to the next state s'. The portfolio value is given by

$$v = p^\top h + b,$$

which is the sum of the market value of all held stocks and the remaining cash balance. Accordingly, the immediate reward is measured by the change in portfolio value between two consecutive time steps. The policy, denoted by $\pi(s)$, represents the trading strategy followed by the agent at state s.

The action-value function, denoted by $Q_\pi(s, a)$, represents the expected cumulative reward obtained by taking action a at state s and then following policy π thereafter.

To describe the trading process explicitly over time, we introduce a time index $t = 0, 1, \ldots, T$. Suppose the initial cash available for investment is $C \in \mathbb{R}_+$, and the investment universe contains n stocks. At each time step t, for each stock $i = 1, \ldots, n$, the agent observes the stock price p_{ti}, the current holding h_{ti}, and the remaining cash balance b_t. Short selling is not allowed. The actions the agent is allowed to take are buying or selling $a_{ti} > 0$ shares or doing nothing. Let $\mathbb{S}$ and $\mathbb{A}$ denote the state and action spaces, then at each time t, a state can be written as

$$\begin{aligned} s_t &= (p_{t1}, \ldots, p_{tn}, h_{t1}, \ldots, h_{tn}, b_t) \in \mathbb{R}_+^n \times \mathbb{R}_+^n \times \mathbb{R}_+, \\ a_t &= (a_{t1}, \ldots, a_{tn}) \in A(s_t). \end{aligned}$$

where $A(s_t)$ is the set of the all feasible actions at the state s_t such that the remaining cash is nonnegative at state s_{t+1}. The reward function at the state-action pair $r(s_t, a_t, s_{t+1})$ is then the profits or loss of taking action a_t at state s_t and reaching state s_{t+1}. The portfolio value at time t is then

$$v_t = (p_{t1}, \ldots, p_{td})^\top (h_{t1}, \ldots, h_{td}) + b_t.$$

Therefore, the immediate reward can be written as

$$r(s_t, a_t, s_{t+1}) = v_{t+1} - v_t.$$

The objective is to find an optimal deterministic policy $\pi \in \Pi, \pi(s_t) = a_t$ that maximizes the portfolio value at a target time T in the future, equivalently maximizing the total rewards $\sum_{t=1}^{T-1} r(s_t, a_t, s_{t+1})$. The expected reward of taking action a_t at state s_t is the current reward plus the discounted expected reward at the next state:

$$V_\pi(s_{t+1}) = \mathbb{E}_{a_{t+1} \sim \pi(s_{t+1})} \left[Q_\pi(s_{t+1}, a_{t+1}) \right].$$

$$Q^\pi(s_t, a_t) = \mathbb{E}_\pi \left[r(s_t, a_t, s_{t+1}) + \gamma V^\pi(s_{t+1}) \mid s_t, a_t \right].$$

where γ is the discount factor. The optimization problem is then equivalent to maximizing $Q_\pi(s_t, a_t)$.

3 Deep Deterministic Policy Gradient (DDPG)

3.1 Network Architecture

Figure 1 illustrates the actor–critic architecture of the DDPG model used in this project. The actor network, denoted by $\mu(s|\theta^\mu)$, takes the current state s as input and outputs an action a. The output action corresponds to the trading decision of the agent. An exploration noise term $\mathcal{N}$ is added to the actor output for better action exploration. The critic network, denoted by $Q(s, a|\theta^Q)$, then receives both the current state and the selected action as inputs and outputs the value of the action under that state. Here θ^μ, θ^Q are the networks parameters. The value of action under that state evaluates the quality of taking action a under state s.

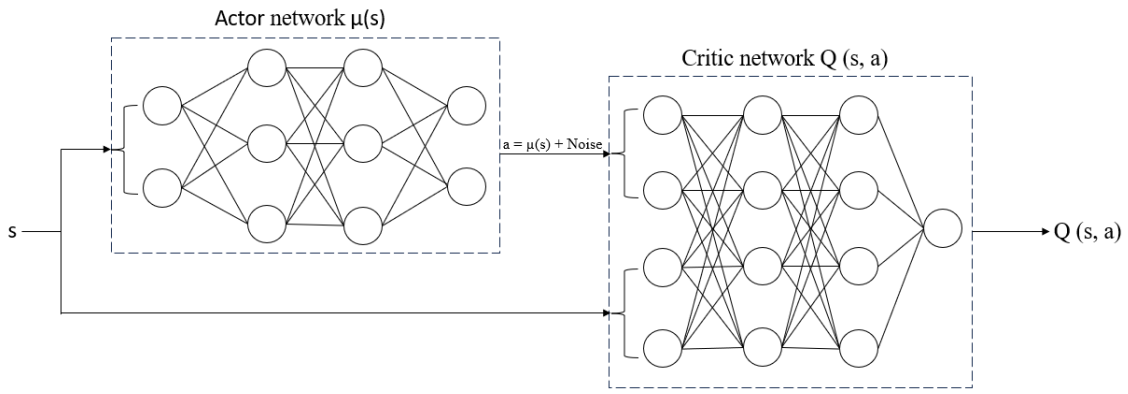


Figure 1: Learning network architecture

3.2 Replay Buffer

To stabilize training, DDPG employs a replay buffer $\mathcal{R}$, which stores the transition tuple

$$(s_t,\, a_t,\, s_{t+1},\, r_t)$$

at each time step t, after the agent takes action a_t at state s_t and transitions to the next state s_{t+1}. A mini-batch of N transitions

$$\{(s_i,\, a_i,\, r_i,\, s_i + 1)\}_{i=1}^{N}$$

is drawn at random from $\mathcal{R}$ to update the actor and critic networks. This random sampling effectively reduces the correlation between experience samples.

3.3 DDPG Algorithm

1. The DDPG algorithm in Liu's paper is specified for the MDP for stock trading. In general, the DDPG algorithm combines the frame work of policy gradient with neural networks as function approximator and Q-learning, which uses greedy action a_{t+1} that maximizes $Q(x_{t+1}, a_{t+1})$ for state x_{t+1}. To be more specific:

2. The actor policy is updated by using the sampled policy gradient, while the critic is updated by minimizing the loss. Suppose at time t, the N sample transitions are drawn from $\mathcal{R}$. Define

$$y_i = r_i + \gamma Q'(s_{i+1}, \mu'(s_{i+1} \mid \theta^{\mu'}) \mid \theta^{Q'}), \quad i = 1, \ldots, N,$$

where μ' and Q' are the target actor and target critic networks, created by copying the actor and critic networks respectively. The critic network is then updated by minimizing the expected difference $L(\theta^Q)$ defined as the expected difference between the outputs of the target critic network Q' and the critic network Q, i.e.,

$$L(\theta^Q) = \mathbb{E}_{s_t, a_t, r_t, s_{t+1} \sim \text{buffer}} \left[\left(y_i - Q(s_t, a_t \mid \theta^Q) \right)^2 \right]. \tag{1}$$

Since a finite number of N sample transitions are drawn from the replay buffer $\mathcal{R}$, the expectation can be approximated by the empirical average, and the loss function 1 can be written as

$$L(\theta^Q) = \frac{1}{N} \sum_{i=1}^{N} \left(y_i - Q(s_i, a_i \mid \theta^Q) \right)^2. \tag{2}$$

The actor network is then updated using the sampled policy gradient, which can be rewritten as

$$\nabla_{\theta^\mu} J \approx \frac{1}{N} \sum_{i=1}^{N} \left[\nabla_a Q(s_i, a \mid \theta^Q) \big|_{a=\mu(s_i)} \nabla_{\theta^\mu} \mu(s_i \mid \theta^\mu) \right]. \tag{3}$$

Following the update of the actor and critic networks using transitions sampled from the replay buffer, the target actor and target critic networks are updated according to:

$$\theta^{Q'} \leftarrow \tau\theta^{Q} + (1-\tau)\theta^{Q'}, \tag{4}$$

$$\theta^{\mu'} \leftarrow \tau\theta^{\mu} + (1-\tau)\theta^{\mu'}, \tag{5}$$

where $\tau \ll 1$ is the soft update rate, ensuring that the target networks track the learned networks slowly for stable training.

4 Case Study

For the case study we replicate the DDPG method and compare with the numerical results in Liu's paper, thus we will use the same experimental settings as Liu whenever they are specified in the paper.

4.1 Numerical experiment setting

The data used are the daily adjusted prices of the 30 stocks of the Dow Jones Industrial Average (DJIA) over the period from January 2, 2009 to September 28, 2018. The tickers included in the dataset are MMM, AMZN, AXP, AMGN, AAPL, BA, CAT, CVX, CSCO, KO, DIS, GS, HD, HON, IBM, JNJ, JPM, MCD, MRK, MSFT, NKE, NVDA, PG, CRM, SHW, TRV, UNH, VZ, V, and WMT. The initial cash for the investment is 10000 and transaction cost is assumed to be 0. In the original paper, the data are divided into training set, validation set and testing set. The training set spans from January 2009 to January 2015 and is used for model training, the validations set is the whole year of 2015, while the test set covers the period from January 2016 to September 2018 and is used for out-of-sample evaluation of all strategies. Validation set is often used to check whether results are theoretically accurate before passing to unseen data, however under the context of stock trading there is no existing theory for comparison, we combine the validation set into the training set.

There are multiple parameters for the actor and critic networks setup:

- The number of layers, dimensions and the choices of activation function;
- discount rate γ and learning rate τ;
- size of the minibatch;
- number of episodes;
- the process which the random noise follows.

None of these parameters are specified in the paper, thus we choose the classical setups. Each network has three layers with 64 sigmoids and all activation functions are linear. The minibatch size is set to $N = 64$, and the exploration noise follows a Gaussian process with standard deviation $\sigma = 0.05$ and mean 0. The actor and critic networks are trained using learning rates of 10^{-4} and 10^{-3}, respectively. The replay buffer capacity is set to 100000 transitions. We attempted to tune the parameters for γ and τ using different parameter settings. Specifically, we test $\gamma = 0.9, 0.93, 0.95, 0.97, 0.99$ and $\tau = 0.001, 0.005$. For each parameter setting, we run the DDPG model 10 times using 100 episodes, record the 10 results, and report their average. While the outputs are unstable, we do notice that $\gamma = 0.99, \tau = 0.001$ always lead to validation results that outperform other combinations. This observation also matches the parameters specified in the paper that introduces the DDPG algorithm. Thus we fix $\gamma = 0.99, \tau = 0.001$ and train the model using 1000 episodes. After that, we show the best-performing in-sample and out-of-sample results obtained among the repeated runs with $M = 100$, using $\gamma = 0.99$ and $\tau = 0.005$.

Portfolio performance is evaluated using several summary statistics computed from the portfolio value series. The initial portfolio value is recorded as the starting wealth, and the final portfolio

value measures the terminal wealth at the end of the investment horizon. Daily returns are computed as

$$r_t = \frac{V_t - V_{t-1}}{V_{t-1}}, \tag{6}$$

where V_t denotes the portfolio value at time t. The annualized return is calculated as the mean daily return multiplied by 252, while the annualized volatility is computed as the standard deviation of daily returns multiplied by $\sqrt{252}$. The Sharpe ratio is then defined as

$$\frac{r_{\text{ann}} - r_f}{\sigma_{\text{ann}}}, \tag{7}$$

where r_f is the risk-free rate. The risk-free rate is also not mentioned in the paper, and we assume it is equal to 1.5%; and in fact, our computation of the mean-variance portfolio using this r_f gives the same result as in the paper. To evaluate downside risk, we also compute the drawdown series, defined as the percentage decline of the portfolio value from its running peak. Based on this series, we report both the maximum drawdown, which captures the largest peak-to-trough loss over the sample period, and the average drawdown, which measures the mean drawdown level across time.

5 Result

5.1 Out of sample results under different parameters

Table 2 presents the out-of-sample performance comparison when $\gamma = 0.9$ and $\tau = 0.001, 0.005$. The results show that the DDPG strategy achieves final portfolio values around \$15,000 to \$17,000 across repeated runs. When $\tau = 0.001$, the average final portfolio value is \$15,714.71, with an average annualized return of 17.24% and an average Sharpe ratio of 1.2451. When $\tau = 0.005$, the average final portfolio value is slightly lower at \$15,659.70, with an average annualized return of 17.07% and an average Sharpe ratio of 1.2769. Overall, the two settings produce similar out-of-sample performance. The setting $\tau = 0.001$ gives a slightly higher average final portfolio value and annualized return, while $\tau = 0.005$ gives a slightly higher Sharpe ratio. This indicates that, when $\gamma = 0.9$, changing τ from 0.001 to 0.005 does not lead to a large difference in the average out-of-sample results.

Experiment	Final Portfolio Value	Annualized Return	Annualized Std.Error	Sharpe Ratio	MaxDD	AVGDD
1	15080.4980	15.77%	12.31%	1.1587	0.1146	0.0250
2	17194.5703	20.72%	13.50%	1.4236	0.1191	0.0200
3	15768.5430	17.48%	12.92%	1.2362	0.1191	0.0275
4	16822.3652	19.84%	12.95%	1.4162	0.1053	0.0219
5	13872.0039	12.70%	12.21%	0.9177	0.1202	0.0304
6	14773.5566	14.96%	11.84%	1.1369	0.1293	0.0270
7	15611.8232	17.12%	12.99%	1.2027	0.1203	0.0236
8	17070.3711	20.35%	12.74%	1.4802	0.0997	0.0199
9	15553.1465	16.90%	12.37%	1.2454	0.1110	0.0247
10	15400.2051	16.52%	12.18%	1.2332	0.1076	0.0221
AVG	15714.7083	17.24%	12.60%	1.2451	0.1146	0.0242

gamma = 0.9, tau = 0.001

Experiment	Final Portfolio Value	Annualized Return	Annualized Std.Error	Sharpe Ratio	MaxDD	AVGDD
1	16365.5625	18.96%	13.88%	1.2586	0.1229	0.0290
2	16889.0410	19.93%	12.46%	1.4790	0.1034	0.0210
3	16601.0176	19.26%	12.18%	1.4586	0.1080	0.0201
4	15184.4326	16.08%	12.77%	1.1412	0.1249	0.0304
5	13925.4102	12.89%	12.58%	0.9057	0.1155	0.0286
6	15080.9131	15.71%	11.83%	1.2010	0.1136	0.0256
7	16428.0410	18.87%	12.09%	1.4366	0.1203	0.0243
8	15720.4736	17.17%	11.31%	1.3859	0.1120	0.0203
9	14423.4453	13.96%	10.76%	1.1579	0.1215	0.0290
10	15978.6611	17.87%	12.17%	1.3448	0.1097	0.0235
AVG	15659.6998	17.07%	12.20%	1.2769	0.1152	0.0252

gamma = 0.9, tau = 0.005

Figure 2: Out-of-sample portfolio value Results under $\gamma = 0.9$, $\tau = 0.001, 0.005$

Table 3 presents the out-of-sample performance comparison when $\gamma = 0.93$ and $\tau = 0.001, 0.005$. The results show that the DDPG strategy achieves final portfolio values around \$14,000 to \$17,000 across repeated runs.

Experiment	Final Portfolio Value	Annualized Return	Annualized Std.Error	Sharpe Ratio	MaxDD	AVGDD
1	16086.5879	18.18%	12.69%	1.3143	0.1069	0.0215
2	16558.9199	19.19%	12.34%	1.4334	0.1157	0.0249
3	16801.8691	19.70%	12.19%	1.4930	0.1120	0.0180
4	15230.2021	16.09%	11.97%	1.2190	0.0995	0.0226
5	16206.6113	18.59%	13.76%	1.2417	0.1360	0.0254
6	14230.3262	13.59%	11.86%	1.0200	0.1160	0.0277
7	16279.9736	18.48%	11.58%	1.4667	0.1342	0.0230
8	15208.9043	16.01%	11.72%	1.2383	0.1341	0.0299
9	15733.0723	17.34%	12.47%	1.2697	0.1058	0.0236
10	17095.0938	20.34%	12.26%	1.5375	0.1100	0.0205
AVG	15943.1561	17.75%	12.28%	1.3234	0.1170	0.0237

gamma = 0.93, tau = 0.001

Experiment	Final Portfolio Value	Annualized Return	Annualized Std.Error	Sharpe Ratio	MaxDD	AVGDD
1	16246.8672	18.55%	12.75%	1.3372	0.1221	0.0235
2	15075.8174	15.79%	12.57%	1.1366	0.1101	0.0269
3	16908.1523	20.08%	13.34%	1.3932	0.1048	0.0230
4	17158.9492	20.48%	12.26%	1.5478	0.1132	0.0209
5	17388.4688	20.98%	12.34%	1.5788	0.1123	0.0163
6	14860.3691	15.29%	12.80%	1.0777	0.1108	0.0241
7	13795.9961	12.53%	12.47%	0.8851	0.1200	0.0328
8	15881.9668	17.72%	12.74%	1.2729	0.1243	0.0223
9	16335.6006	18.66%	12.05%	1.4237	0.1063	0.0197
10	16082.4180	18.15%	12.54%	1.3280	0.1135	0.0228
AVG	15973.4605	17.82%	12.59%	1.2981	0.1137	0.0232

gamma = 0.93, tau = 0.005

Figure 3: Out-of-sample portfolio value Results under $\gamma = 0.93$, $\tau = 0.001, 0.005$

Figure 4 presents the out-of-sample performance comparison when $\gamma = 0.95$ and $\tau = 0.001, 0.005$. The results show that the DDPG strategy achieves final portfolio values around \$14,000 to \$19,000 across repeated runs.

Experiment	Final Portfolio Value	Annualized Return	Annualized Std.Error	Sharpe Ratio	MaxDD	AVGDD
1	17419.6750	20.98%	11.84%	1.6453	0.1037	0.0143
2	16216.1016	18.47%	12.69%	1.3371	0.1091	0.0256
3	15497.5020	16.72%	11.91%	1.2772	0.1076	0.0243
4	15327.3994	16.45%	12.98%	1.1516	0.1386	0.0244
5	16894.7852	19.90%	12.10%	1.5197	0.1156	0.0235
6	14876.7646	15.34%	12.84%	1.0775	0.1312	0.0299
7	16075.6631	18.06%	11.93%	1.3882	0.1131	0.0208
8	16322.8877	18.61%	11.86%	1.4419	0.1170	0.0244
9	16204.3428	18.38%	12.19%	1.3853	0.1031	0.0222
10	16539.4219	19.10%	11.96%	1.4713	0.1206	0.0238
AVG	16137.4543	18.20%	12.23%	1.3695	0.1160	0.0233

gamma = 0.95, tau = 0.001

Experiment	Final Portfolio Value	Annualized Return	Annualized Std.Error	Sharpe Ratio	MaxDD	AVGDD
1	15711.4346	17.31%	12.68%	1.2470	0.1194	0.0262
2	17511.9004	21.29%	12.75%	1.5517	0.1246	0.0248
3	19029.3828	24.37%	13.06%	1.7507	0.1199	0.0158
4	14810.2666	15.04%	11.78%	1.1494	0.1035	0.0225
5	17088.9707	20.36%	12.50%	1.5094	0.1152	0.0217
6	18166.3340	22.63%	12.76%	1.6562	0.1068	0.0173
7	14234.4551	13.59%	11.78%	1.0271	0.1306	0.0325
8	15907.9424	17.83%	13.13%	1.2431	0.1173	0.0289
9	17141.6328	20.56%	13.14%	1.4508	0.1131	0.0241
10	15051.0234	15.71%	12.45%	1.1415	0.1070	0.0245
AVG	16465.3343	18.87%	12.60%	1.3727	0.1157	0.0238

gamma = 0.95, tau = 0.005

Figure 4: Out-of-sample portfolio value Results under $\gamma = 0.95$, $\tau = 0.001, 0.005$

Figure 5 presents the out-of-sample performance comparison when $\gamma = 0.97$ and $\tau = 0.001, 0.005$. The results show that the DDPG strategy achieves final portfolio values around \$14,000 to \$19,000 across repeated runs.

Experiment	Final Portfolio Value	Annualized Return	Annualized Std.Error	Sharpe Ratio	MaxDD	AVGDD
1	15432.2217	16.73%	13.22%	1.1522	0.1164	0.0251
2	15976.9766	17.80%	11.67%	1.3974	0.1158	0.0208
3	15937.6338	17.88%	12.99%	1.2608	0.1055	0.0223
4	13339.3213	11.29%	12.35%	0.7929	0.1388	0.0388
5	16072.2207	18.01%	11.56%	1.4275	0.1227	0.0233
6	15816.5762	17.48%	12.10%	1.3214	0.1011	0.0201
7	16089.8311	18.24%	13.13%	1.2751	0.1068	0.0219
8	18565.2461	23.42%	12.75%	1.7199	0.1141	0.0199
9	15459.0781	16.59%	11.61%	1.2999	0.1264	0.0259
10	16546.5566	19.03%	11.25%	1.5590	0.1148	0.0194
AVG	15923.5662	17.65%	12.26%	1.3206	0.1162	0.0237

gamma = 0.97, tau = 0.001

Experiment	Final Portfolio Value	Annualized Return	Annualized Std.Error	Sharpe Ratio	MaxDD	AVGDD
1	16402.9980	18.77%	11.72%	1.4731	0.1116	0.0219
2	14478.4707	14.21%	11.71%	1.0855	0.1211	0.0272
3	14800.7939	15.10%	12.42%	1.0950	0.1015	0.0236
4	15913.5420	17.86%	13.31%	1.2291	0.1170	0.0273
5	18967.3594	24.22%	12.84%	1.7690	0.1068	0.0176
6	15637.8613	17.07%	12.11%	1.2853	0.1022	0.0194
7	16283.3457	18.50%	11.73%	1.4496	0.1094	0.0188
8	16401.8047	18.93%	13.06%	1.3350	0.1086	0.0231
9	16831.3945	19.73%	11.86%	1.5365	0.0978	0.0181
10	16769.8672	19.82%	13.62%	1.3450	0.1266	0.0204
AVG	16248.7438	18.42%	12.44%	1.3603	0.1103	0.0218

gamma = 0.97, tau = 0.005

Figure 5: Out-of-sample portfolio value Results under $\gamma = 0.97$, $\tau = 0.001, 0.005$

Figure 6 presents the out-of-sample performance comparison when $\gamma = 0.99$ and $\tau = 0.001, 0.005$. The results show that the DDPG strategy achieves final portfolio values around \$14,000 to \$19,000 across repeated runs.

Experiment	Final Portfolio Value	Annualized Return	Annualized Std.Error	Sharpe Ratio	MaxDD	AVGDD
1	15423.4590	16.59%	12.28%	1.2285	0.1160	0.0232
2	15667.3311	17.03%	11.16%	1.3919	0.1124	0.0210
3	15996.5283	17.90%	12.10%	1.3551	0.1061	0.0211
4	15782.8516	17.38%	11.92%	1.3329	0.1077	0.0222
5	14976.4492	15.57%	12.73%	1.1051	0.1194	0.0245
6	13979.4688	12.93%	11.76%	0.9722	0.1161	0.0287
7	15173.2764	16.01%	12.43%	1.1670	0.1089	0.0258
8	16611.3457	19.23%	11.67%	1.5183	0.1186	0.0222
9	15603.2744	16.92%	11.49%	1.3417	0.1211	0.0241
10	15735.0469	17.44%	13.26%	1.2022	0.1265	0.0230
AVG	15494.9031	16.70%	12.08%	1.2615	0.1153	0.0236

gamma = 0.99, tau = 0.001

Experiment	Final Portfolio Value	Annualized Return	Annualized Std.Error	Sharpe Ratio	MaxDD	AVGDD
1	15500.8994	16.69%	11.65%	1.3038	0.1122	0.0255
2	16511.2012	19.17%	12.99%	1.3605	0.1134	0.0243
3	17174.3438	20.68%	13.53%	1.4170	0.1453	0.0200
4	16490.9492	18.94%	11.48%	1.5186	0.1157	0.0194
5	15068.8115	15.69%	11.93%	1.1895	0.1139	0.0228
6	15790.3223	17.36%	11.57%	1.3705	0.1146	0.0220
7	16386.1523	18.83%	12.50%	1.3867	0.1008	0.0203
8	14718.2520	14.93%	12.69%	1.0585	0.1160	0.0289
9	16835.8242	19.78%	12.20%	1.4981	0.1087	0.0217
10	16645.8047	19.30%	11.67%	1.5257	0.1153	0.0187
AVG	16112.2561	18.14%	12.22%	1.3629	0.1156	0.0224

gamma = 0.99, tau = 0.005

Figure 6: Out-of-sample portfolio value Results under $\gamma = 0.99$, $\tau = 0.001, 0.005$

The results above show that the DDPG strategy is unstable across repeated runs and different parameter settings. Even when the same values of γ and τ are used, the final portfolio value, annualized return, Sharpe ratio, maximum drawdown, and average drawdown vary across the 10 repeated experiments. However, among the parameter settings tested, the combination $\gamma = 0.95$ and $\tau = 0.005$ gives the best average out-of-sample performance, with the highest average final portfolio value and annualized return. Therefore, we use this parameter setting to further train the DDPG model with $M = 1000$ episodes.

5.2 Out of sample results under parameters M= 1000

Figure 7 shows the results of five experiments with $M = 1000$. The final portfolio values are around \$15,000. However, the results are still not very stable, although they are more stable than the results with $M = 100$. After this comparison, the best result among the five runs is selected and compared with the result reported in the paper.

Experiment	Final Portfolio Value	Annualized Return	Annualized Std.Error	Sharpe Ratio	MaxDD	AVGDD
1	15316.2295	16.31%	12.11%	1.2226	0.1182	0.0251
2	15500.0400	16.70%	11.74%	1.2953	0.1245	0.0251
3	14472.682,	14.28%	12.41%	1.0296	0.1193	0.0303
4	16198.8750	18.42%	12.60%	1.3431	0.1090	0.0225
5	14838.2810	15.06%	11.32%	1.1979	0.1086	0.0216
AVG	15265.2215	16.15%	12.04%	1.2177	0.1159	0.0249

M = 1000, gamma = 0.95, tau = 0.005

Figure 7: Out-of-sample portfolio value Results under $M = 1000$, $\gamma = 0.95$, $\tau = 0.005$

5.3 Out-of-sample result

Table 1 presents the out-of-sample performance comparison for the best-performing run. The results show that the DDPG strategy achieves higher performance than the two benchmark strategies across the main performance measures. Since the paper does not specify all parameter settings, and because the DDPG training process is affected by randomness, the exact result reported in the paper cannot be reproduced.

Table 1: Out-of-Sample Performance Comparison

	DDPG (paper)	DDPG (ours)	Mean-Variance	DJIA
Initial Value	10000.00	10000.00	10000.00	10000.00
Final Value	19791.00	16933.28	14382.96	15428.54
Annualized Return	25.87%	20.05%	14.20%	16.53%
Annualized Std. Error	13.62%	12.65%	10.50%	11.71%
Sharpe Ratio	1.79	1.47	1.21	1.28
Max Drawdown	NA	10.85%	15.00%	11.58%
AVG Drawdown	NA	2.29%	3.83%	2.35%

Figure 8 shows the DDPG strategy significantly outperforms DJIA and the mean-variance portfolio allocation. While the DDPG strategy achieves substantially higher returns in the in-sample period, its out-of-sample performance is much closer to the benchmark strategies. This reduction in performance gap suggests that part of the strong in-sample results may be attributed to overfitting to historical data. In particular, although DDPG still maintains slightly higher returns and risk-adjusted performance, the advantage is less pronounced compared to the in-sample results. We also observe that the DDPG strategy produces different results each time the code is executed. This is due to the stochastic nature of the training process, including random initialization of network parameters, random minibatch sampling from the replay buffer, and exploration noise added to the actor policy. As a result, the learned policy is not exactly the same across different runs, which leads to variation in portfolio performance.

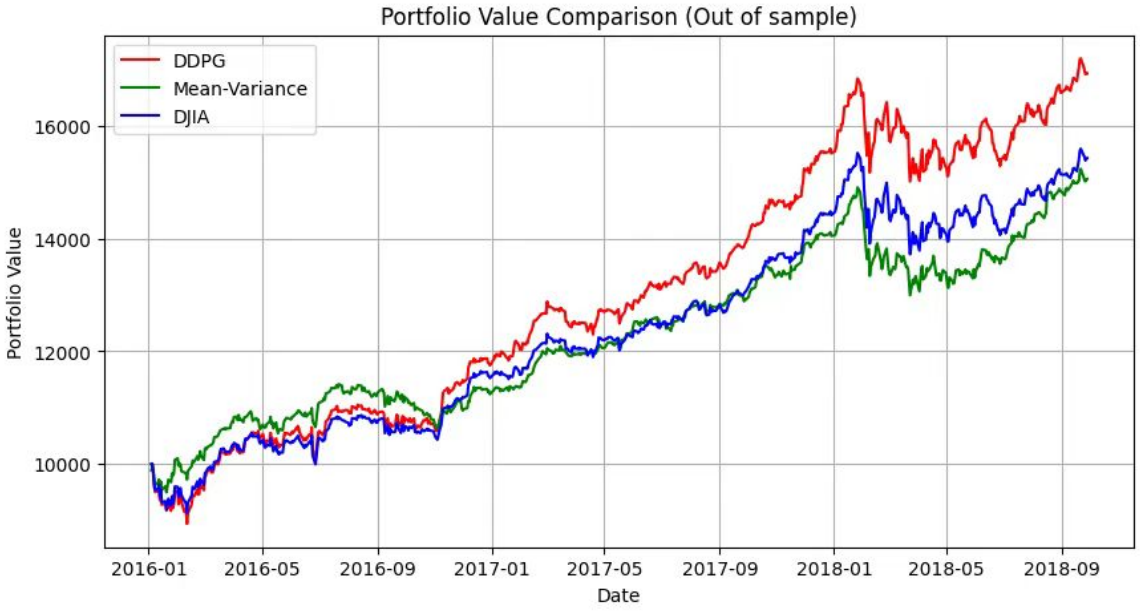


Figure 8: Out-of-sample portfolio value curves of our DDPG scheme, the mean-variance portfolio allocation strategy, and the Dow Jones Industrial Average (DJIA), with an initial portfolio value of $10,000.

Figure 9 illustrates the drawdown dynamics of the DDPG strategy, the mean-variance portfolio, and the DJIA benchmark for the out-of-sample periods. The mean-variance strategy exhibits a sustained and relatively high drawdown during certain intervals, particularly around 2017–2018, indicating weaker adaptability to changing market conditions. In comparison, the DDPG strategy maintains lower and more controlled drawdowns over most of the period, with smaller and shorter-lived spikes. The DJIA shows moderate drawdowns, generally lying between the other two strategies. The major drawdown periods can also be related to historical market events. The drawdown in early 2016 is associated with the global stock market selloff [6], while the drawdown in mid-to-late 2018 is associated with the U.S.–China trade war [7] and related market uncertainty.

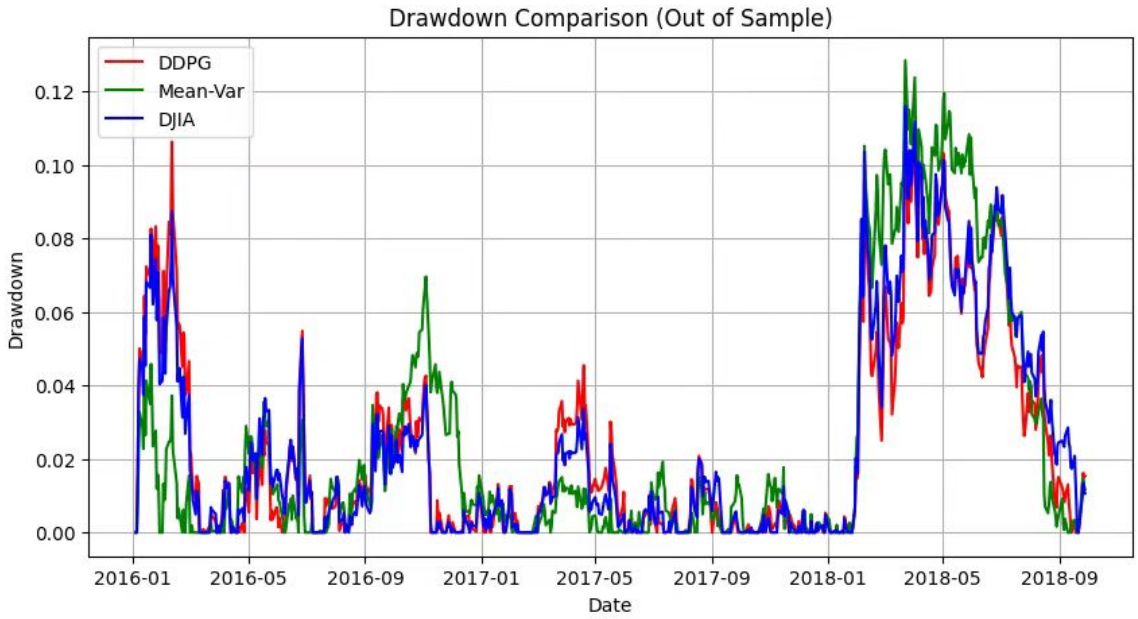


Figure 9: Out-of-sample drawdown curves of our DDPG scheme, the mean-variance portfolio allocation strategy, and the Dow Jones Industrial Average (DJIA), with an initial portfolio value of $10,000.

6 Conclusion and Future Work

This report investigated the application of the Deep Deterministic Policy Gradient (DDPG) algorithm to portfolio optimization using the 30 constituent stocks of the DJIA. The DDPG strategy was compared with two benchmark strategies: the mean-variance portfolio and the passive DJIA index.

The empirical results show that, under certain parameter settings, the DDPG-based strategy achieves slightly better out-of-sample performance than both benchmarks. Specifically, it has a higher final portfolio value, annualized return, and Sharpe ratio compared with the mean-variance portfolio and the DJIA index.

However, we could not determine whether out sample performs are stable. We referred to the original paper introducing DDPG [2], where 2.5 million timesteps were used to train a dataset with state space and action space dimensions of at most 37 and 12. Our state space dimension is 61 and action space dimension is 30, and 1000 episodes (about 1.7 million timesteps) is insufficient in comparison and much more episodes might be needed. However we are in capable in testing larger episodes as 1000 episodes already cost 3 hours for training. Future work can first focus on improving the stability, we might be able to achieve this by decreasing the data complexity by considering fewer stocks. After determining when model converges, more hyperparameters can be tuned. In addition, Ornstein–Uhlenbeck (OU) process noise can be used instead of Gaussian noise. The action space can be changed to portfolio weights.

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